

Mannat Vikramaditya Jain

Columbia Engineering Profile

EDUCATION

- **Columbia University** New York, NY
B.S. Materials Science and Engineering, Eggleston Scholar *co' 2030 (on leave for MATS)*

RESEARCH EXPERIENCE

- **MATS 11.0 Scholar** Berkeley, CA
Fourth Eon Biosecurity *Sept. 2026*
 - Researching learning dynamics in biological foundation models.
 - Distilling pathogenicity detection from Evo-2 into concept-bottlenecked students for improved DNA synthesis screening.
- **MARS Fellow** Oxford, UK / Remote
Goodfire *Jul. 2026 – Present*
 - Developing weight-based interpretability techniques to improve data sampling for atomistic simulations.
 - Mentored by Thomas Dooms (Goodfire Life Sciences) and Ward Gauderis.
 - Received \$10k BlueDot rapid grant to extend research.
- **Gang Lab, Columbia Chemical Engineering** New York, NY
Research Intern *Sept. 2025 – May 2026*
 - Built a microscopy data-analysis pipeline in Python to quantify CRISPR-Cas9 DNA origami degradation kinetics.
 - Second author on paper using enzyme kinetics to organize dynamic DNA nanostructures (in preparation).
 - Sole undergraduate working on the project.
- **Duke Department of Bioinformatics** Durham, NC / Remote
Research Intern *May 2024 – Jul. 2025*
 - Finetuned ESM-2 on REBASE for synthetic restriction enzyme design.
 - Abstract and poster accepted to DNA31 (ENS de Lyon), awarded NSF travel grant to present.
 - Advised by Dr. David Page, Chair of Biostatistics & Bioinformatics at Duke, with international team from Lodz and Giessen Universities.
- **MIT Laboratory for Computational Physiology** Remote
Research Intern *May 2024 – Jul. 2025*
 - Identified ethnic biases in ICU documentation patterns across 53,111 stays in Midwest hospitals.
 - Analyzed the MIMIC-III, MIMIC-IV, and eICU Collaborative Healthcare databases.
 - Published two abstracts (second author, coauthor) in *Critical Care Medicine* (IF 7.7), presented at SCCM Congress 2025.

AI SAFETY

- **Columbia AI Alignment Club** New York, NY
Director *Jan. 2026 – Aug. 2026*
 - Managed \$30,000+ in funding from Kairos and ran logistics for 6 fellowships.
 - Organized speaker events by researchers at OpenAI and Redwood Research.
 - Authoring an AIXBio curriculum for a fall reading group; led CAIAC team and first-authored submission for Apart Research AIXBio hackathon (awarded AI Biosecurity Tools track spotlight).

HONORS & AWARDS

- 2026, Contrary Research Fellow
- 2026, CAMBRIA (ARENA curriculum)
- 2025, National Merit Scholar
- 2025, Ellison Scholar Semifinalist
- 2023-24, 2x Presidential Volunteer Service Award, Gold
- 2023-24, 3x Grand Award, NYSSEF & LISEF
- 2024, USAID International Science Champion
- 2023, Congressional App Challenge, 1st Place